

12.421 / 12.621 Lecture 10

Topics covered

- Review of Earth's atmosphere structure
- Brief discussion of temperature lapse rate and pressure and density profiles
- Absorption and emission by gases in the atmosphere

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Interaction of EM radiation and atmospheres

- Virtually all remote sensing applications need to contend w/ atmosphere, whether the atmosphere is signal or noise
- As we saw, the transparency of the atmosphere varies w/ EM frequency from virtually opaque to virtually transparent in a way that may appear at first glance to be quasi-random
- To understand how the atmosphere can be observed w/ remote sensing and can influence terrestrial observations, we start w/ basic definitions of the atmosphere

Standard Atmosphere of Earth

- Earth's atmosphere is made up of 5 primary layers. Layers boundaries are defined by changes in thermal lapse rate. From top to bottom, the layers are
 - Exosphere
 - Altitude range ~ 700 Km (AMSL) - $\sim 10^4$ Km
 - Extremely low density
 - bottom is defined by "exobase" $\rightarrow$ also known as thermopause $\sim$ orbital altitude of many Earth-observing satellites
 - low density + high altitude $\rightarrow$ negligible for most RS applications
 - Thermosphere
 - Altitude range ~ 80 Km - ~ 700 Km
 - low density (many satellites and ISS orbit here)
 - temperature increases w/ elevation (temperature inversion) from ~ 190 K to ~ 330 K (in International Standard Atmosphere)
 - top is defined by exobase (a.k.a. thermopause)
 - bottom defined by mesopause
 - Importantly for RS applications, ionosphere is located at the bottom of Thermosphere

Atmo at Earth (cont.)

• Thermosphere (cont.)

- ionosphere is the ionized layer of Earth's atmo; plays a key role in EM wave propagation from satellites and radio propagation on Earth
- ionosphere is a layer of free electrons; electrons are stripped from atoms and molecules of the atmo by ultraviolet and higher frequency radiation from the Sun; electrons remain free for some time due to low density of thermosphere; eventually, electron will be recaptured by a positive ion (process called recombination); low density enables rate of release of electrons $\approx$ rate of recombination w/ positive ions $\Rightarrow$ persistent plasma (ionosphere) w/ temporal variability
- ionosphere is dispersive at microwave frequencies, enables detailed studies using dual-frequency GPS
- Strongest influence on RS is at very high and low (microwave and lower) frequencies

• Mesosphere

- Altitude range $\approx$ 50 km - 80 km
- Temperature decreases w/ elevation (it lapse rate) from $\approx$ 270 K at the base down to $\approx$ 190 K at the mesopause
- top defined by mesopause
- base defined by stratopause
- can allow for formation of ice-crystal clouds, called noctilucent clouds

Atmo of Earth (cont.)

• Stratosphere

- Altitude range $\approx$ 12 km (tropopause) - $\approx$ 50 km (stratopause)
- Temperature increases w/ altitude (i.e. - lapse rate or temp. inversion) from $\approx$ 213 K at the tropopause to $\approx$ 270 K at the stratopause
- Temperature inversion caused by absorption of ultraviolet radiation by the ozone layer
- Very dry, though some water vapor produced in situ by photochemical oxidation of methane
- No weather (bc no turbulence), but polar stratospheric or nacreous clouds can form in winter at high latitudes

• Troposphere ('tropo' is Greek for 'turn')

- Altitude range from Earth's surface to $\approx$ 12 km (tropopause)
- Upper altitude varies as a function of latitude and other factors, such as weather
- In general, temperature decreases w/ altitude due to heating from Earth's surface; temperature inversions can and do occur
- Accounts for vast majority of total mass of atmosphere $\Rightarrow$ most dense part of atmo
- Holds virtually all atmospheric moisture $\Rightarrow$ ^{can have} major impact on most forms of RS
- All weather and virtually all clouds are found in tropo $\Rightarrow$ object of many RS observations and noise source for others

* To a good approximation in most RS applications, we can consider the atmosphere to be the ionosphere and the troposphere

Properties of the troposphereTemperature

- Common temperature profiles are defined by International, U.S., or other Standard Atmospheres
- In the U.S. and International Standard Atmospheres, temperature is linear w/ elevation in the troposphere
- The slope is known as the "temperature lapse rate", a function of moisture content.
- For a dry atmosphere, 1st Law of Thermodynamics for adiabatic process given as

$$p c_p dT = dp$$

$T \equiv$ temperature

$p \equiv$ pressure

$\rho_{\text{air}} \equiv$ mass density of dry air

$c_p \equiv$ specific heat (const. pressure)

Assume hydrostatic

$$\Rightarrow dp = -\rho g dz$$

$g \equiv$ grav. accel.

Dry adiabatic lapse rate is

$$a_d = -\frac{dT}{dz} = \frac{g}{c_p} \approx \frac{9.81 \text{ m/s}^2}{1000 \text{ J/kg}\cdot\text{K}} \approx 9.8 \frac{\text{K}}{\text{km}} \quad (\text{a constant})$$

- Moisture in the atmo complicates the lapse rate; must account for condensation. (involves release of latent heat)

$\Rightarrow a_w \equiv$ moist adiabatic lapse rate

$$= f(T, p) \quad \text{most strongly dependent on } T$$

- Standard atmosphere does not explicitly account for moisture, but sets constant (within tropo) temperature lapse

rate $\boxed{a \equiv 6.5 \frac{\text{K}}{\text{km}}}$

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Properties of troposphere

Pressure

- Can be shown that pressure varies w/ altitude z as

$$P = P_0 \left(1 - \frac{a z}{T_0} \right)^{\frac{g M}{a R}}$$

$a \equiv$ temperature lapse rate

$T_0 \equiv$ sea-level standard temperature

$M \equiv$ molar mass of dry air

$R \equiv$ universal gas const

- Assuming $a = a_d = \frac{g}{c_p} \equiv$ dry adiabatic lapse rate

$$P = P_0 \left(1 - \frac{g z}{c_p T_0} \right)^{\frac{c_p M}{R}}$$

- Given $c_p \approx 10^3 \frac{J}{kg K}$, $M \approx 2.9 \times 10^{-2} \frac{kg}{mol}$, and $R = 8.314 \frac{J}{mol K}$,
binomial approx gives

$P \approx P_0 e^{-z/H_p}$	$H_p = \frac{g M}{T_0 R} \approx 8.4 \text{ km} \overset{\text{pressure}}{\equiv} \text{scale height of atm}$
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Density

- The identical argument can be made for density of dry air as for pressure

$\Rightarrow \rho_{air} = \rho_0 e^{-z/H_p}$ where $H_p \approx H_\rho$ and $\rho_0 \equiv$ density of dry air at sea level

- Water vapor density varies in space and time, standard atm gives

$\rho_v = \rho_{v0} e^{-z/H_v}$ where $\rho_{v0} \approx 7.72 \times 10^{-3} \frac{kg}{m^3}$
and $H_v \approx 2 \text{ to } 2.5 \text{ km}$

Absorption and Emission by gases

(very high-level)

- Brief overview of the fundamental concepts of absorption and emission by atmospheric gases
- Total ^{internal} energy U of isolated molecule consists of 3 energy states

$$U = U_e + U_v + U_r$$

$U_e \equiv$ electronic energy $\rightarrow$ electron energy levels in individual atoms

$U_v \equiv$ vibrational energy $\rightarrow$ change of position between two atoms

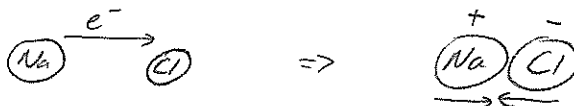
$U_r \equiv$ rotational energy $\rightarrow$ rotation of molecule

- Energy states are quantized $\Rightarrow$ take on discrete values specified by one or more quantum numbers
- Let us consider each energy state at a very high level
 - For illustrative purposes, let us consider diatomic molecules

: A note on bonding:

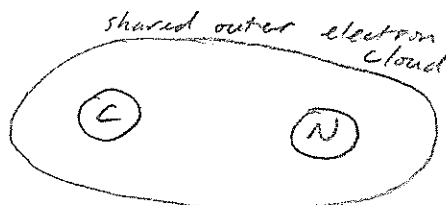
- Ionic bonding - electrostatic attraction between atoms - one atom loses an ^{outer} electron (+ charge) and the other gains electron (- charge)

E.g. sodium chloride (sodium loses electron)



- Covalent bonding - atoms share outer electrons

E.g.



- With either type of bond, we may expect a gradient in electric charge $\Rightarrow$ polarization, or electric dipole $\Rightarrow$ applied electric field can cause rotation

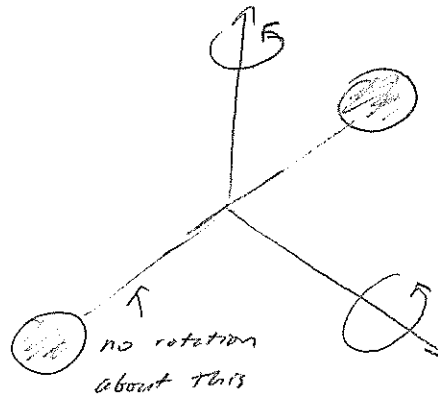
Absorption and emission by gases

• Energy states (cont.)

- Different energy states

• Rotational energy

↳ Our diatomic molecule can rotate "end over end" but can not rotate about axis connecting atoms



(can't rotate relative to one another and synchronous rotation is meaningless $\rightarrow$ vrt no energy)

↳ laws of quantum mechanics require rotations only at discrete rates or frequencies (not continuous as in classical mechanics)

↳ But like classical mechanics, higher rotation rates imply higher energy

↳ discrete rotation rates and rates $\leftrightarrow$ energy $\Rightarrow$ described by "rotation levels" $\rightarrow$ energy state U_r

↳ Key feature is that rotational transitions can occur by absorbing or emitting photons, * but can only 'jump' from one rotation level to another and only one at a time *

↳ To move from one rotation level to another, molecule must absorb a photon w/ frequency equal to that of the next rotation level (frequency)
 $\Rightarrow$ to spin faster, molecules must absorb increasingly higher frequency EM radiation

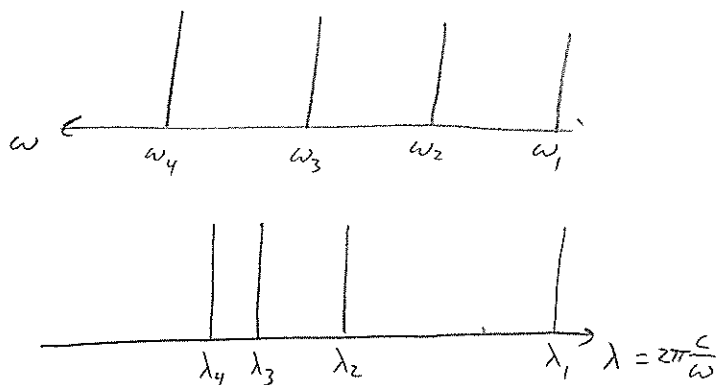
Absorption and emission by gases

• Energy states (cont).

- Rotational energy (cont)

conversely,
 ↳ to slow down, a molecule must emit EM radiation
 w/ frequency equal to the rotation level

↳ rotation levels are equally spaced in energy, and hence frequency, therefore absorption lines are equally spaced in frequency but not wave length



$$↳ U_r = \frac{\hbar^2 \sqrt{l(l+1)}}{2} \quad \text{where } \hbar \equiv \text{reduced Planck constant } (\hbar = \frac{h}{2\pi})$$

$l \equiv \text{nonnegative integer called the rotational quantum number}$

- Vibrational energy U_v

↳ dipoles can be 'smushed' together along the line joining the atoms

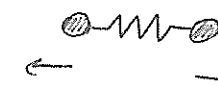
↳ basic process

1)



positive side of dipole is pushed by EM radiation and negative side is pulled, causing the atomic nuclei to move closer to one another

2)



nuclei are both positive charged $\Rightarrow$ repel (analogous to a spring connecting two weights)

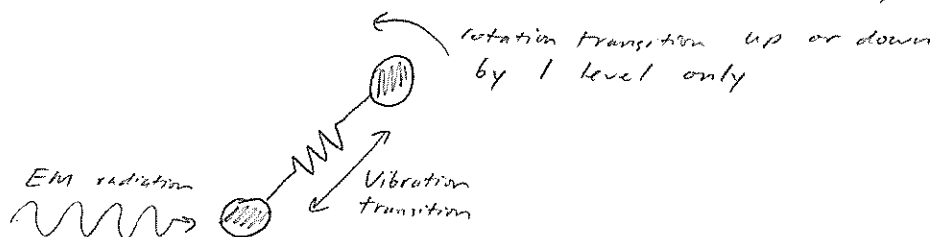
$\Rightarrow$ EM radiation pushes nuclei together, like charges repel $\Rightarrow$ vibration

Absorption and emission by gases

• Energy states (cont.)

- Vibration (cont.)

- ↳ frequency of vibrations are quantized $\rightarrow$ vibration levels
- ↳ When a molecule absorbs EM radiation of the frequency associated w/ a higher vibration level, a vibration transition occurs
- ↳ * However * vibration transitions are always accompanied by a rotation transition $\rightarrow$ as we saw in the previous section, rotation transitions involve jumps of one and only one level (can be up or down)



- ↳ Since each vibration level can be associated w/ several rotation (sub-) levels, there are many possible vibration transitions within a population
- ↳ All vibration transitions require slightly different energy levels, results in a series of closely spaced absorption lines that converge at short wavelengths (i.e. higher frequencies) $\Rightarrow$ absorption bands
- ↳ Key point is that a range of higher frequency EM can be absorbed by molecules in the atmo $\rightarrow$ visible red to thermal infrared portion of EM spectrum
- ↳
$$\nu = \left(n_v + \frac{1}{2}\right) h\nu_0$$
 where n_v is an integer called the vibrational quantum number